Wei-Cheng Lee

weicheng.frank.lee@gmail.com · website · DBLP · OpenReview

Ph.D. student in Computer Science, RL / optimization theory

Research Interests

I develop practical optimization algorithms with strong theoretical guarantees, with a focus on online learning, bandits, and reinforcement learning. I am also interested in statistical learning theory, fundamental limits in stochastic optimization, and AI for math.

Education

King Abdullah University of Science and Technology (KAUST) — Dec. 2025 – Present
Ph.D. Student in Computer Science — *Thuwal, Saudi Arabia*
- Advisor: Francesco Orabona.
- Research focus: practical optimization algorithms with theoretical guarantees, online learning, bandits, reinforcement learning, statistical learning theory, and fundamental limits in stochastic optimization.

King Abdullah University of Science and Technology (KAUST) — Aug. 2024 – Dec. 2025
M.Sc. in Computer Science — *Thuwal, Saudi Arabia*
- Continued into the KAUST Computer Science Ph.D. program.
- Selected coursework: online learning, high-performance computing, numerical methods for stochastic differential equations, computing systems and concurrency.

National Taiwan University — Sep. 2012 – Jan. 2018
B.Sc. in Computer Science, Minor in Mathematics — *Taipei, Taiwan*
- Selected coursework: algorithm design and analysis, real analysis, abstract algebra, linear algebra, probability theory, numerical methods.

Publications

A Single Stepsize Suffices for Unprojected Linear TD(0): Simultaneous Robust and Fast Rates via Polyak-Ruppert Averaging — COLT 2026
Wei-Cheng Lee, Francesco Orabona.

New Lower Bounds for Stochastic Non-Convex Optimization through Divergence Decomposition — COLT 2025
El Mehdi Saad, **Wei-Cheng Lee**, Francesco Orabona.

MixLasso: Generalized Mixed Regression via Convex Atomic-Norm Regularization — NeurIPS 2018
Ian E.H. Yen, **Wei-Cheng Lee**, Kai Zhong, Sung-En Chang, Pradeep Ravikumar, Shou-De Lin.

Latent Feature Lasso — ICML 2017
Ian E.H. Yen, **Wei-Cheng Lee**, Sung-En Chang, Arun S. Suggala, Shou-De Lin, Pradeep Ravikumar.

A Best-of-Both-Worlds Proof for Tsallis-INF without Fenchel Conjugates — arXiv
Wei-Cheng Lee, Francesco Orabona.

A Robust $\widetilde{\mathcal{O}}(1/\sqrt{T})$ Rate for Unprojected TD Learning with Linear Function Approximation — arXiv
Wei-Cheng Lee, Francesco Orabona.

Research Experience

Graduate Research, King Abdullah University of Science and Technology (KAUST) — Aug. 2024 – Present
M.Sc./Ph.D. Student in Computer Science — *Thuwal, Saudi Arabia*
- Proved finite-time high-probability guarantees for unprojected linear TD(0) under Markovian sampling, showing that a single stepsize schedule yields bounded iterates and both robust curvature-free and fast curvature-dependent rates via Polyak-Ruppert averaging.
- Established query-complexity lower bounds for first-order stochastic non-convex optimization under structural conditions such as quasar-convexity, quadratic growth, and restricted secant inequalities, using divergence decomposition to reduce optimization to hard function-identification problems.
- Developed a simplified primal analysis of best-of-both-worlds guarantees for Tsallis-INF in multi-armed bandits, replacing Fenchel-conjugate machinery with online-convex-optimization tools.

Institute of Information Science, Academia Sinica — Feb. 2022 – Mar. 2024
Research Assistant — *Taipei, Taiwan*
- Proved regret guarantees for online reinforcement learning in infinite-horizon average-reward MDPs with full-information feedback, developing a new entropic-linear-program analysis for Hedge-style behavior across stochastic and adversarial regimes.

Graduate Research, National Taiwan University — Feb. 2018 – Jun. 2021
M.Sc. Student — *Taipei, Taiwan*
- Derived algorithms for log-loss function classes, with emphasis on online portfolio selection.

Engineering Experience

Saudi Aramco, Research & Development Center Jun. 2025 – Jul. 2025
Graduate Intern *Dhahran, Saudi Arabia*

- Built and solved a gas-oil separation plant network optimization model as a mixed-integer nonlinear program in Pyomo.
- Extended optimization-modeling workflows by integrating JAX-based derivative computation with finite-difference validation for gradients, Jacobians, and Hessians.

Graduate Institute of Biomedical Electronics and Bioinformatics, NTU Aug. 2021 – Jan. 2022
Research Assistant *Taipei, Taiwan*

- Applied BERT-based language models to transform clinical cases into ICD-10 codes for medical insurance applications.

Teaching

Reinforcement Learning, KAUST Spring 2026, Spring 2025
Teaching Assistant

Optimization Algorithms, National Taiwan University Fall 2019, Fall 2018
Teaching Assistant

Probability, National Taiwan University Spring 2018, Spring 2015
Teaching Assistant

Honors

Presidential Award, National Taiwan University Spring 2014, Fall 2012
Awarded to the top 5% of students each semester

Technical Stack

Programming: Python, JAX, PyTorch, C/C++, MATLAB, Julia, Go, JavaScript.